

University Roll Number	Question Paper Code
J 2 5 0 3 5 4 0 2 9	14125105

JSS MAHAVIDYAPEETHA
JSS UNIVERSITY, NOIDA
MCA

First Year Theory Examination 2025-26

Course Name:	OPERATING SYSTEM WITH LINUX	Max. Marks:	60
Course Code:	25MCA 140 / JCA 105	Time:	3 Hrs

Section A		2 x 5 = 10	BL	CO
Attempt ALL Questions in brief.				
1.	a. Differentiate between Time-Sharing OS and Real-Time OS.		2	CO1
	b. Define internal and external fragmentation.		1	CO2
	c. Describe directory structure in an operating system?		2	CO3
	d. What is difference between seek time and Latency in disk scheduling?		2	CO4
	e. List two disk-related commands in Linux		2	CO5

Section B																	
Attempt ANY ONE of the following Question.			7 x 1 = 07	BL	CO												
a.	Describe the major components of an Operating System and explain the services provided by each component.			2	CO1												
OR																	
2.	Consider the following processes: Apply Preemptive SJF (Shortest Remaining Time First) calculate: 1. Gantt chart 2. Completion time of each process 3. Waiting time 4. Turnaround time 5. Average waiting & turnaround time			3	CO1												
b.	Process Arrival Time Burst Time <table border="0" style="width: 100%;"> <tr> <td>P1</td> <td>0</td> <td>8</td> </tr> <tr> <td>P2</td> <td>1</td> <td>4</td> </tr> <tr> <td>P3</td> <td>2</td> <td>9</td> </tr> <tr> <td>P4</td> <td>3</td> <td>5</td> </tr> </table>			P1	0	8	P2	1	4	P3	2	9	P4	3	5		
P1	0	8															
P2	1	4															
P3	2	9															
P4	3	5															

Attempt ANY ONE of the following Question.		7 x 1 = 07	BL	CO
	a. Explain virtual memory. Describe demand paging and factors affecting its performance.		2	CO2
OR				
	b. Considering a system with five processes P0 through P4 and three resources of type A, B, C. Resource type A has 10 instances, B has 5 instances and type C has 7 instances. Suppose at time t0 following snapshot of the system has been taken:		4	CO2

Process	Allocation			Max			Available		
	A	B	C	A	B	C	A	B	C
P ₀	0	1	0	7	5	3	3	3	2
P ₁	2	0	0	3	2	2			
P ₂	3	0	2	9	0	2			
P ₃	2	1	1	2	2	2			
P ₄	0	0	2	4	3	3			

Calculate the Safe Sequence.

Attempt ANY ONE of the following Question.			7 x 1 = 07	BL	CO
4.	a.	Explain file sharing and file protection mechanisms in operating systems.	2	CO3	
OR					
	b.	Describe different file allocation methods (Contiguous, Linked, Indexed) with advantages and disadvantages.	2	CO3	

Attempt ANY ONE of the following Question.			7 x 1 = 07	BL	CO
5.	a.	Discuss the need and working of RAID levels in mass storage systems.	2	CO4	
OR					
	b.	Suppose the order of request is- (82,170,43,140,24,15,190) and current position of Read/Write head is: 50 (200) Calculate Total overhead movement using SCAN and C-SCAN	3	CO4	

Attempt ANY ONE of the following Question.			7 x 1 = 07	BL	CO
	a.	Describe the history, salient features, and architecture of Linux in detail.	2	CO5	
OR					
	b.	Write a shell script for any three of the following tasks and explain them: Display current date and time Display the calendar Accept user input using read Perform arithmetic using bc Display system information using uname	3	CO5	

Section C

Attempt ANY THREE of the following Questions			5 x 3 = 15	BL	CO
		Differentiate between Monolithic Kernel and Microkernel	2	CO1	
		Calculate the Page Fault by FIFO page replacement Algorithm on following page reference string 5, 0, 1, 0, 2, 3, 0, 2, 4, 3, 3, 2, 0, 2, 1, 2, 7, 0, 1. (Frame = 3)	4	CO2	
		Explain the concept of free-space management. Mention any two techniques.	2	CO3	
		Discuss and Compare primary, secondary, and tertiary storage based on speed and cost.	2	CO4	
		Explain any three basic Linux commands: date, who, tty, cal, echo.	2	CO5	
